

Non-Factorisability and the Emergence of Heisenberg Structure from Admissibility Constraints

Jérôme Beau
Independent Researcher
`jerome.beau@cosmochrony.org`

2026

Abstract

We prove that the admissibility axioms of the Cosmochrony framework, together with Born–Infeld parity, uniquely determine the algebraic structure of the admissible fibre F_n . The central result is that any structure of observable directions that is finite, minimal, and genuinely non-pre-resolved is necessarily of Heisenberg type: the symmetry group of F_n is isomorphic to $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for a prime q , and F_n is isomorphic to the Weil representation V_ρ on $L^2(\mathbb{Z}/q\mathbb{Z})$. The proof proceeds through a single structural lemma — admissible non-factorisability — which is a direct formalisation of Axiom A3, and requires no input beyond A1–A3, Born–Infeld parity, and classical classification results. As a corollary, the Heisenberg uncertainty principle is derived as a structural property of the non-resolved proto-state, not as an independent postulate about observables. This paper provides the detailed proof of the result stated in sketch form in the companion paper on admissible non-injective transitions [2].

Keywords. Admissibility; non-factorisability; Heisenberg group; Weil representation; Stone–von Neumann theorem; Born–Infeld parity; proto-state; uncertainty principle; canonical quantisation; emergent quantum structure.

Contents

1	Introduction	3
2	Setup and Notation	4
2.1	The four axioms	4
2.2	The admissible fibre and its generators	4
2.3	The Weil representation	5
3	The Admissible Non-Factorisability Lemma	6
4	Proof of the Main Theorem	7
4.1	Step 1: Non-trivial commutator	7
4.2	Step 2: Central commutator	7
4.3	Step 3: Classification as the Heisenberg algebra	8
4.4	Step 4: The order of Z is prime	9
4.5	Step 5: Identification $F_n \simeq V_\rho$	9
5	Corollaries	11
5.1	The uncertainty principle as a structural corollary	11
5.2	Canonical quantisation as structural necessity	11
5.3	Relation to the O-series	11
6	Discussion	13
6.1	What the theorem establishes	13
6.2	The single non-trivial input	13
6.3	Open direction: the large- q limit	13

1 Introduction

The Cosmochrony programme [1] derives quantum structure from a minimal relational foundation, without postulating a background spacetime, a Hilbert space, or an external substrate. Its primitive is the local structure of admissible transitions between observable states, governed by four axioms.

The companion paper [2] establishes that Axiom A3 — admissibility as non-premature selection — forces phase coherence and identifies the admissible fibre F_n as a precursor of Hilbert structure. It states, as a structural remark, that F_n is concretely realised as the Weil representation V_ρ of the discrete Heisenberg group $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$.

The present paper provides the detailed proof of this identification. The main result, Theorem 4.1, is:

Under A1–A3 and Born–Infeld parity, the symmetry group of F_n is isomorphic to $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for some prime q , and $F_n \simeq V_\rho$.

The proof rests on a single structural lemma (Lemma 3.1), which formalises the prohibition on premature resolution encoded in A3. Once this lemma is established, the rest of the argument is a chain of classification steps: non-trivial commutator, central commutator, Heisenberg algebra, prime order of the centre, and Stone–von Neumann identification.

Relation to companion papers

This paper is a structural companion to [2], which states Axioms A1–A4 and derives their consequences. The non-abelian exclusion and quaternionic minimality used in Remark 5.4 are established in detail in [5]. The Born–Infeld parity involution $\sigma: c \mapsto q - c$ is derived in [4]. The non-injectivity of Π_n as a structural necessity is proved in [3].

Organisation

Section 2 recalls the axiomatic setup and fixes notation. Section 3 establishes the admissible non-factorisability lemma. Section 4 proves the main theorem in five steps. Section 5 derives the uncertainty principle and other corollaries. Section 6 discusses the conceptual significance and open directions.

2 Setup and Notation

We recall the axioms and fix the notation used throughout. The full axiomatic development is given in [2].

2.1 The four axioms

- A1 — Local projective admissibility.** For any observable state O_{n-1} , there exists a non-empty set F_n of admissible successor directions. The set F_n is defined by internal structural constraints (Born–Infeld bound, spectral envelope, symmetry requirements) and does not presuppose an ambient configuration space.
- A2 — Structural non-injectivity.** Distinct admissible directions from O_{n-1} may lead to the same resolved successor state O_n .
- A3 — Admissibility as non-premature selection.** An admissible transition preserves the full multiplicity of open directions until projection locking occurs. It does not select among them prematurely — that is, before the Born–Infeld saturation threshold is reached.
- A4 — Projection locking.** Resolution occurs only when continuous Born–Infeld saturation meets the discrete support of the observable image of the projection. In the spectral admissibility realisation, this support is the shell decomposition of the observable space.

2.2 The admissible fibre and its generators

We assume throughout that the admissible fibre F_n is finite-dimensional. This follows from the Born–Infeld admissibility bound [4]: the constraint $A_n \leq c_\chi/\sqrt{\lambda_n}$ bounds the effective amplitude of each mode, and only finitely many modes satisfy it at any given cascade depth n .

By A1, the admissible fibre F_n is a non-empty set of admissible directions from O_{n-1} . It carries a group of symmetry transformations, whose generators are constrained by the Born–Infeld bound and the spectral envelope. We denote the symmetry group of F_n by G_n ; its action on F_n is by definition faithful.

Born–Infeld parity [4] asserts that the BI action satisfies $S_{\text{BI}}[\chi] = S_{\text{BI}}[-\chi]$, which forces the involution $\sigma: c \mapsto q - c$ on the central character of the Weil representation. At the level of generators, this means: for every generator X of G_n , there is a conjugate generator $\sigma(X)$, the image of X under σ . By A1, no generator beyond the conjugate pair $(X, \sigma(X))$ is required to generate the admissible symmetry group: the admissibility constraints — BI bound, spectral envelope, and symmetry requirements — are entirely encoded by

the pair, and any additional independent generator would constitute structure not forced by these constraints. The pair $(X, \sigma(X))$ therefore forms the minimal admissible generating set of G_n .

Definition 2.1 (Stable decomposition). *A direct sum decomposition $F_n = F^{(1)} \oplus F^{(2)}$ is called stable (under admissible generators) if both $F^{(1)}$ and $F^{(2)}$ are invariant under the action of every generator of G_n .*

Definition 2.2 (Admissible resolution). *A direction $f \in F_n$ is admissibly resolved if there exists a stable decomposition $F_n = F^{(1)} \oplus F^{(2)}$ such that $f \in F^{(1)}$ or $f \in F^{(2)}$, i.e. f is assigned to a definite sector prior to any Born–Infeld saturation event.*

2.3 The Weil representation

For a prime q and a non-zero character $c \in (\mathbb{Z}/q\mathbb{Z})^\times$, the discrete Heisenberg group $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z}) = \{(a, b, \gamma) : a, b, \gamma \in \mathbb{Z}/q\mathbb{Z}\}$ with multiplication $(a, b, \gamma)(a', b', \gamma') = (a + a', b + b', \gamma + \gamma' + ab')$ admits an irreducible representation of dimension q , the Weil representation ρ_c , acting on $V_\rho = L^2(\mathbb{Z}/q\mathbb{Z}) \cong \mathbb{C}^q$ by

$$(\rho_c(a, b, \gamma) f)(x) = e^{2\pi i c(\gamma + bx)/q} f(x - a). \quad (1)$$

The central element $(0, 0, 1)$ acts as multiplication by $e^{2\pi i c/q} \neq 1$ (since $c \neq 0$), so the central character is non-trivial.

3 The Admissible Non-Factorisability Lemma

The following lemma is the structural core of the proof. It is a direct formalisation of Axiom A3.

Lemma 3.1 (Admissible non-factorisability). *Under A1–A3, every stable decomposition of F_n is trivial, i.e. one of $F^{(1)}$ and $F^{(2)}$ must be $\{0\}$. In particular, the action of G_n on F_n is irreducible.*

Proof. Suppose $F_n = F^{(1)} \oplus F^{(2)}$ is a non-trivial stable decomposition, with $F^{(1)} \neq \{0\}$ and $F^{(2)} \neq \{0\}$. By Definition 2.1, both summands are invariant under every generator of G_n . Since the generators of G_n are the constitutive elements of the admissibility structure of F_n (A1), the decomposition is distinguished by the admissibility structure itself: the sector to which a direction f belongs ($F^{(1)}$ or $F^{(2)}$) is determined by an invariant internal to F_n .

This constitutes an admissible resolution in the sense of Definition 2.2: each open direction carries a definite sectorial label, accessible within the admissibility structure of F_n , prior to any Born–Infeld saturation event.

A3 forbids premature selection among open directions: the transition must remain structurally neutral among all elements of F_n until projection locking forces resolution. An admissible resolution prior to locking is precisely what A3 prohibits. Therefore the non-trivial stable decomposition is forbidden.

Irreducibility follows immediately: any proper G_n -invariant subspace $W \subsetneq F_n$ with $W \neq \{0\}$ would yield a non-trivial stable decomposition $F_n = W \oplus W^\perp$ (since G_n acts unitarily, the orthogonal complement of a G_n -invariant subspace is also G_n -invariant). This contradicts the non-factorisability just proved. $\square$

Remark 3.2 (Lemma 3.1 as a restatement of A3). *The content of Lemma 3.1 may be expressed as:*

$$\text{proto-state genuinely non-resolved} \iff F_n \text{ carries no admissible sectorial decomposition.} \quad (2)$$

In this sense, the lemma is not an additional hypothesis: it is A3 rephrased in the language of representation theory.

Remark 3.3 (Unitarity of the G_n action). *The action of G_n on F_n is unitary. This follows from the BI bound: admissible transformations preserve the norm structure of F_n , since the BI saturation threshold is defined in terms of mode amplitudes $\|f\|$ [4]. Unitarity is used in the proof above to ensure that the orthogonal complement of an invariant subspace is again invariant.*

4 Proof of the Main Theorem

Theorem 4.1 (A1–A3 and BI parity force Heisenberg structure). *Under A1–A3 and Born–Infeld parity, the symmetry group G_n of the admissible fibre F_n is isomorphic to $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for some prime q . Consequently, F_n is isomorphic to the Weil representation V_ρ on $L^2(\mathbb{Z}/q\mathbb{Z})$.*

The proof proceeds in five steps.

4.1 Step 1: Non-trivial commutator

Proposition 4.2. *Under A1–A3, the conjugate generators X and $\sigma(X)$ satisfy $[X, \sigma(X)] \neq 0$.*

Proof. Suppose $[X, \sigma(X)] = 0$. Then X and $\sigma(X)$ are commuting unitary operators on the finite-dimensional space F_n . By the spectral theorem for commuting normal operators, they admit a simultaneous eigenspace decomposition:

$$F_n = \bigoplus_{\lambda} F_{\lambda}, \quad (3)$$

where F_{λ} is the common eigenspace for the pair of eigenvalues $(x_{\lambda}, y_{\lambda}) \in \text{U}(1)^2$ of $(X, \sigma(X))$. This decomposition is non-trivial (since X and $\sigma(X)$ are generators of G_n , they are not scalar multiples of the identity by faithfulness of the action). Each summand F_{λ} is invariant under X and $\sigma(X)$, and hence, since $(X, \sigma(X))$ is the minimal generating set of G_n by A1, under all of G_n . Therefore the decomposition is stable. By Lemma 3.1, a non-trivial stable decomposition is forbidden. Contradiction. Therefore $[X, \sigma(X)] \neq 0$. $\square$

4.2 Step 2: Central commutator

Proposition 4.3. *Let $Z := [X, \sigma(X)]$. Under A1, Z is central in G_n .*

Proof. Suppose Z is not central. Then there exists a generator $Y \in G_n$ such that $[Z, Y] \neq 0$. The group generated by $(X, \sigma(X), Z, Y)$ is then strictly richer than the group generated by the conjugate pair $(X, \sigma(X))$ alone: the pair (Z, Y) with $[Z, Y] \neq 0$ constitutes an independent non-abelian substructure not generated by the commutator $Z = [X, \sigma(X)]$. Concretely, the iterated commutator $[[X, \sigma(X)], Y]$ is non-trivial and independent of the generators already present, so the derived series of G_n does not terminate at the subgroup $\langle Z \rangle$: the group is not nilpotent of class 2 over the conjugate pair. This produces a strictly larger derived subalgebra than that generated by the pair $(X, \sigma(X))$ alone, contradicting the assumption that all admissible relations are generated by this pair under A1.

By A1, F_n is generated minimally by the conjugate pair $(X, \sigma(X))$: the admissibility constraints force no generator beyond those required by

Born–Infeld parity, and the algebra of G_n is entirely determined by the BI-admissible structure of the pair. An independent non-abelian sub-structure beyond what the pair generates would constitute an internal structure of F_n not determined by the admissibility constraints, contradicting the minimality of A1. Therefore Z is central in G_n . $\square$

Remark 4.4 (Nilpotency class 2). *Propositions 4.2 and 4.3 together imply that G_n is nilpotent of class exactly 2: the commutator subgroup is $[G_n, G_n] = \langle Z \rangle$, which is central and non-trivial. This is precisely the nilpotency class of the Heisenberg group.*

4.3 Step 3: Classification as the Heisenberg algebra

Proposition 4.5. *The group G_n is isomorphic to a Heisenberg group.*

Proof. By Propositions 4.2 and 4.3, G_n is a group with:

- (i) a generating pair $(X, \sigma(X))$ related by the involution σ ;
- (ii) a non-trivial central commutator $Z = [X, \sigma(X)]$;
- (iii) no independent algebraic relation beyond those forced by minimality (A1).

Condition (iii) is made precise as follows: minimality (A1) excludes any additional independent commutation relation beyond those generated by the conjugate pair $(X, \sigma(X))$, so the only non-trivial relation in G_n is $[X, \sigma(X)] = Z$ with Z central. Conditions (i)–(iii) mean that G_n is a non-split central extension of the abelian group $\langle X \rangle \times \langle \sigma(X) \rangle$ by the cyclic group $\langle Z \rangle$, with the extension class determined by the commutator pairing $(X, \sigma(X)) \mapsto Z$. Since no additional relation is present, this is the *universal* central extension of $\langle X \rangle \times \langle \sigma(X) \rangle$ with the given commutator.

The group cohomology classification of central extensions of $\mathbb{Z}/q\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$ by $\mathbb{Z}/q\mathbb{Z}$ is given by $H^2(\mathbb{Z}/q\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}, \mathbb{Z}/q\mathbb{Z}) \cong \mathbb{Z}/q\mathbb{Z}$ [8], generated by the class corresponding to the standard symplectic commutator $[(a, 0), (0, b)] = ab \bmod q$. Up to automorphism of $(\mathbb{Z}/q\mathbb{Z})^2$, all non-degenerate alternating bilinear forms are equivalent (since q is prime, hence $(\mathbb{Z}/q\mathbb{Z})^2$ is a two-dimensional symplectic space over $\mathbb{F}_q$, and $\text{Sp}(2, \mathbb{F}_q)$ acts transitively on non-degenerate forms), yielding a unique non-split extension class. The unique non-split extension (up to isomorphism) is the Heisenberg group:

$$G_n \cong \text{Heis}_3(\mathbb{Z}/q\mathbb{Z}) = \langle X, \sigma(X), Z \mid [X, \sigma(X)] = Z, Z \text{ central}, X^q = \sigma(X)^q = Z^q = 1 \rangle. \quad (4)$$

The commutator defines a non-degenerate alternating bilinear form $\omega: \langle X \rangle \times \langle \sigma(X) \rangle \rightarrow \langle Z \rangle$ by $\omega(X^a, \sigma(X)^b) = Z^{ab}$ on the generator space. Non-degeneracy follows from $Z \neq 0$ and the fact that X and $\sigma(X)$ are independent generators: $\omega(X, \cdot) \neq 0$ and $\omega(\cdot, \sigma(X)) \neq 0$. This form fixes the Heisenberg structure up to isomorphism. $\square$

4.4 Step 4: The order of Z is prime

Proposition 4.6. *The order $|Z|$ of the central generator Z is prime. Hence q in (4) is prime.*

Proof. Write $|Z| = \prod_i p_i^{a_i}$ for the prime factorisation. Suppose $|Z|$ is not prime. Then the centre $\langle Z \rangle \cong \mathbb{Z}/|Z|\mathbb{Z}$ contains at least one proper non-trivial subgroup, say $H = \langle Z^d \rangle$ for some proper divisor $d > 1$ of $|Z|$.

Since H is a subgroup of the centre $Z(G_n)$, it is normal in G_n . The quotient map $\pi: G_n \rightarrow G_n/H$ induces a decomposition of any G_n -representation via the characters of H . More precisely, since H acts as scalars (being central), Schur's lemma implies that any irreducible G_n -representation on which H acts trivially factors through G_n/H , while any representation on which H acts non-trivially decomposes F_n into H -eigenspaces:

$$F_n = \bigoplus_{\chi \in \hat{H}} F_\chi, \quad (5)$$

where $F_\chi = \{v \in F_n : h \cdot v = \chi(h)v \ \forall h \in H\}$ and $\hat{H}$ denotes the character group of H . If $|H| > 1$, this decomposition is non-trivial (there is more than one H -eigenspace, since F_n is not a direct sum of trivial H -representations — Z acts non-trivially on F_n as the commutator of generators).

Each H -eigenspace F_χ is invariant under the full action of G_n : for $g \in G_n$ and $v \in F_\chi$, and $h \in H \leq Z(G_n)$,

$$h \cdot (g \cdot v) = g \cdot (h \cdot v) = \chi(h)(g \cdot v), \quad (6)$$

so $g \cdot v \in F_\chi$. Therefore the decomposition $F_n = \bigoplus_\chi F_\chi$ is stable under all of G_n , hence under admissible generators. By Lemma 3.1, this stable decomposition is forbidden. Therefore $|H| = 1$, i.e. $\langle Z \rangle$ has no proper non-trivial subgroup. A cyclic group with no proper non-trivial subgroup is cyclic of prime order. Hence $|Z| = q$ is prime. $\square$

4.5 Step 5: Identification $F_n \simeq V_\rho$

Proposition 4.7. *F_n is isomorphic to the Weil representation V_ρ of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$.*

Proof. By Propositions 4.5 and 4.6, $G_n \cong \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for a prime q . By Lemma 3.1, F_n is an irreducible G_n -representation. It remains to identify which irreducible representation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ it is.

By the discrete Stone–von Neumann theorem (see e.g. [9]), $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ has, for each non-zero character $c \in (\mathbb{Z}/q\mathbb{Z})^\times$, a unique irreducible representation of dimension q with central character $\chi_c: (0, 0, \gamma) \mapsto e^{2\pi i c \gamma / q}$. All irreducible representations with non-trivial central character are of this form (up to unitary equivalence), and they are all isomorphic to the Weil representation (1). In addition, $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ has q^2 one-dimensional representations (those factoring through the abelianisation), but these are excluded

by irreducibility of F_n together with $q \geq 2$ (since $\dim F_n \geq 2$ by Proposition 4.2: a one-dimensional representation would be abelian, contradicting $[X, \sigma(X)] \neq 0$).

Born–Infeld parity forces $c \neq 0$ [4]: the parity involution $\sigma: c \mapsto q - c$ has no fixed point in $(\mathbb{Z}/q\mathbb{Z})^\times$ (since q is odd prime and $q - c \neq c$ for $c \neq 0$), so any admissible character is non-trivial. Therefore F_n carries a non-trivial central character, and by the Stone–von Neumann theorem, $F_n \simeq V_\rho$. $\square$

Propositions 4.2–4.7 together constitute the proof of Theorem 4.1. $\square$

Remark 4.8 (Summary of the derivation chain). *The logical chain of Theorem 4.1 is:*

A3 $\implies$ *no stable decomposition* (Lem. 3.1) $\implies [X, \sigma(X)] \neq 0$ (Prop. 4.2),

A1 $\implies$ *minimal generative closure* $\implies Z$ *central* (Prop. 4.3),

A1 $\implies$ *no excess relation* $\implies G_n \cong \text{Heis}_3$ (Prop. 4.5),

A3 $\implies$ *irreducibility* $\implies |Z| = q$ *prime* (Prop. 4.6),

BI parity $\implies c \neq 0 \implies F_n \simeq V_\rho$ (Prop. 4.7).

(7)

Every arrow is either a direct consequence of an axiom or a classical algebraic fact. No external input is required beyond A1–A3 and Born–Infeld parity.

5 Corollaries

5.1 The uncertainty principle as a structural corollary

Corollary 5.1 (Heisenberg uncertainty principle). *Under A1–A3 and Born–Infeld parity, conjugate admissible observables X and $\sigma(X)$ satisfy*

$$\Delta X \cdot \Delta \sigma(X) \geq \frac{1}{2} |\langle [X, \sigma(X)] \rangle| = \frac{1}{2} |\langle Z \rangle| > 0. \quad (8)$$

In particular, X and $\sigma(X)$ cannot be simultaneously sharp.

Proof. By Proposition 4.2, $[X, \sigma(X)] = Z \neq 0$. The standard Robertson uncertainty relation for unitary operators on a Hilbert space gives the bound above. The strict positivity follows from $Z \neq 0$. $\square$

Remark 5.2 (Uncertainty as a property of the non-resolved proto-state). *Corollary 5.1 establishes that the Heisenberg uncertainty principle is not a property of the observables per se, but a structural consequence of A3 alone: the prohibition on premature resolution forces $[X, \sigma(X)] \neq 0$, from which the uncertainty bound follows. Equivalently, the uncertainty principle is the algebraic signature of the fact that the proto-state is genuinely non-resolved: if the proto-state were resolved (in the sense of carrying a stable sectorial decomposition), the commutator would be zero and simultaneous sharp values would be possible.*

5.2 Canonical quantisation as structural necessity

Corollary 5.3 (Canonical quantisation). *Under A1–A3 and Born–Infeld parity, the algebra of admissible observables on F_n is the canonical quantisation of the discrete symplectic space $(\mathbb{Z}/q\mathbb{Z})^2$ with symplectic form $\omega(X, \sigma(X)) = Z$.*

Proof. By Theorem 4.1, $G_n \cong \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and $F_n \simeq V_\rho$. The Heisenberg group is, by construction, the canonical quantisation of the symplectic space $((\mathbb{Z}/q\mathbb{Z})^2, \omega)$, in the sense that it is the unique group whose Lie algebra (or discrete analogue) encodes the commutation relation $[X, \sigma(X)] = Z$ with Z central and ω non-degenerate. The Weil representation V_ρ is the unique faithful irreducible representation of this quantisation. $\square$

5.3 Relation to the O-series

Remark 5.4 (Non-abelian support from O23). *Theorem 4.1 is complementary to the result of [5], which establishes via the Hurwitz classification that the admissible neutral sector of F_n is non-abelian (quaternionic). The present paper establishes the group structure of G_n directly from A1–A3 without assuming the Hurwitz classification, using only the commutation argument of Proposition 4.2. The two derivations are therefore independent*

and mutually reinforcing: O_{23} establishes why the neutral sector is quaternionic and three-dimensional; the present paper establishes why the ambient symmetry group is Heisenberg and the representation is the Weil representation.

6 Discussion

6.1 What the theorem establishes

Theorem 4.1 establishes three things simultaneously.

First, it closes the identification $F_n \simeq V_\rho$, which was stated as a structural remark in [2] and as the target of open question Q5 therein. The space in which the proto-state lives between transitions is the Weil representation V_ρ of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, uniquely determined by the axioms. This space is real in the sense of the Cosmochrony framework — it is not spatial, but it is not epistemic either: it is the space of structurally admissible directions, induced by the admissibility constraints themselves.

Second, it derives canonical quantisation as a structural necessity rather than a modelling choice. The Heisenberg commutation relation $[X, \sigma(X)] = Z$ does not enter as a postulate; it is forced by A3 via the non-factorisability lemma. The imaginary unit and the complex Hilbert space structure follow from this relation, not from an independent axiom.

Third, it resolves the open question Q6 of [2]: the Heisenberg uncertainty principle is derived as a corollary of A3 alone, without any assumption about the nature of the observables.

6.2 The single non-trivial input

The proof rests on exactly one non-trivial structural claim: the admissible non-factorisability lemma (Lemma 3.1). All other steps use standard algebraic tools (simultaneous diagonalisation, group cohomology classification, Stone–von Neumann).

Lemma 3.1 itself is a direct formalisation of A3. Its proof requires: (a) unitarity of the G_n action (from the BI bound), (b) the definition of admissible generators as constitutive of the admissibility structure (A1), and (c) the prohibition on premature selection (A3). None of these is new: (a) is established in [4], (b) is the content of A1, and (c) is A3.

6.3 Open direction: the large- q limit

Theorem 4.1 establishes the structure of F_n at finite prime q . The large- q limit — in which $\mathbb{C}^q \rightarrow L^2(\mathbb{R})$ and the Weil representation approaches the metaplectic representation of the real Heisenberg group $\text{Heis}_3(\mathbb{R})$ — is the subject of the companion spectral geometry programme [1]. The emergence of continuous spacetime geometry from the BFS shell structure of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ in the large- q limit (with homogeneous dimension $D = 4$ from the Bass–Guivarc’h theorem) remains an open theorem.

Acknowledgements

The author acknowledges the use of large language models as a supportive tool for refining language, structure, and internal consistency during the development of this manuscript. All conceptual contributions, theoretical choices, and interpretations remain the sole responsibility of the author.

References

- [1] J. Beau, “Cosmochrony: A Pre-Geometric Relational Framework for Emergent Physics,” white paper, Zenodo (2026). doi:10.5281/zenodo.17957509.
- [2] J. Beau, “Admissible Non-Injective Transitions as the Primitive of Physical Description,” Cosmochrony companion paper M, Zenodo (2026). doi:10.5281/zenodo.19616329.
- [3] J. Beau, “Non-Injectivity as a Structural Necessity of Genuine Emergence: A No-Go Theorem for Faithful Emergent Descriptions,” Cosmochrony companion paper L, Zenodo (2026). doi:10.5281/zenodo.19391746.
- [4] J. Beau, “Bounded Relaxation and the Dynamical Selection of Space-time Geometry,” Cosmochrony companion paper C, Zenodo (2026). doi:10.5281/zenodo.18407505.
- [5] J. Beau, “Three Stable Directions from Quaternionic Minimality: Derivation of $\Sigma_c(n_3) = 3$ from Born–Infeld Fibre Admissibility,” O23 spectral admissibility sub-programme paper, Zenodo (2026). doi:10.5281/zenodo.19375136.
- [6] J. Beau, “Quaternionic Rigidity of Admissible Morphisms: Every Admissible $\Phi_{q,\rho}$ Necessarily Factors through $\mathfrak{su}(2)$,” O27 spectral admissibility sub-programme paper, Zenodo (2026). doi:10.5281/zenodo.19489296.
- [7] J. Beau, “From Admissibility to Quantum Structure: Phase Coherence and Correlations from Non-Injective Projection,” Cosmochrony companion paper Q1, Zenodo (2026). doi:10.5281/zenodo.19561060.
- [8] K. S. Brown, *Cohomology of Groups*, Graduate Texts in Mathematics **87**, Springer, New York (1982). doi:10.1007/978-1-4684-9327-6.
- [9] P. Gérardin, “Weil representations associated to finite fields,” *Journal of Algebra* **46**, 54–101 (1977). doi:10.1016/0021-8693(77)90394-5.